

PUMPING LEMMA PRACTICE SHEET AUTOMATA & COMPUTABILITY (CSE331)

Q: Prove they are Non - Regular.

1. $\{w \in \{0,1,2\}^*: 0^n 1^n 2^n \text{ where } n \geq 0\}$
2. $\{w \in \{0,1\}^*: 0^x 1^y 0^z \text{ where } z > x+y \text{ and } x, y \geq 0\}$
3. $\{w \in \{0,1\}^*: \text{where } w \text{ is a palindrome}\}$
4. $\{w \in \{0,1\}^*: \text{where } w \text{ is not a palindrome}\}$
5. $\{w \in \{a,b\}^*: \text{number of } a \text{ in } w \text{ is a prime number}\}$
6. $\{w \in \{0\}^*: 0^{3^n} \text{ where } n \geq 0\} \mid \{w = 1^n: n \text{ is a perfect cube (e.g., } n=1, 8, 27, \dots)\}$
7. $\{w \in \{0,1\}^* \mid ww \text{ where } n \geq 0\}$
8. $\{w \in \{0\}^*: 1^{n^2} \text{ where } n \geq 0\}$
9. $\{w \in \{0,1\}^*: 0^i 1^j \text{ where } i > j\}$
10. $\{w = 1^n: n \text{ is a power of two}\}$
11. $\{ww^R \mid w \in \Sigma^* \text{ and } w^R \text{ means } w \text{ in reverse}\}$
12. $\{w_1 \# w_2 \text{ such that length of } w_1 = \text{length of } w_2\}$
13. $\{w \in \Sigma^* \mid w = 0^i 1^j \text{ where } i < 3j\}$
14. $\{w \in \{0,1\}^*: 0^{2n} 1^n \text{ where } n \geq 0\}$
15. $\{w \in \{0,1\}^*: 0^n 1^n \text{ where } n! = m\}$
16. $\{w \in \{0,1,2,3\}^*: 1^n 0^m 3^n 2^m \text{ where } n, m \geq 0\}$
17. $\{w \in \{0,1,2,3\}^*: 0^a 1^b 1^c 0^d \text{ where } a+b=c+d\}$

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Editing in progress

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